

Revision G Electrical Design

Updated 8 October 2026. This revision removes the external host-reset circuit from Revision F. It retains the Arduino Nano Every, three Serial Wombat PCB0046 HSD V2 boards, proportional valve, command and feedback modules, and local supply capacitors. There is no relay, relay regulator, permanent flow meter, or external temperature sensor.

Status and scope

This is a documented bench prototype, not physically tested hardware. Flashable Nano Every firmware v1 is included and compiled; physical hardware remains unqualified. See [firmware instructions](#). The schematic, connection register and automated checks establish documented connections; they do not establish actual valve behavior, thermal performance or software recovery. Use the same component revisions and check the following before machine installation.

Check	Required evidence
H1 - Machine harness	Identify actual supply, return and G/H/J cavities. The five IN wire names are not Takeuchi connector cavity numbers. Measure key-off behavior.
H2 - Supply and wiring	Existing fuse rating must suit the added wiring. Confirm conductor sizes, voltage under load and shared return connections. No new machine fuse block is specified.
H3 - Purchased parts	Confirm Nano Every, HSD V2, input-board revision, lamp polarity/current.
H4 - Valve interface	Confirm five-wire actuator, signal-common compatibility, command compliance, reported-position behavior and startup/shutdown/USB sequencing.
H5 - Firmware and bench	Bench-test the implemented internal watchdog, closed startup, faults, settings storage, buttons, current/voltage measurements where fitted, enclosure temperature and water operation.

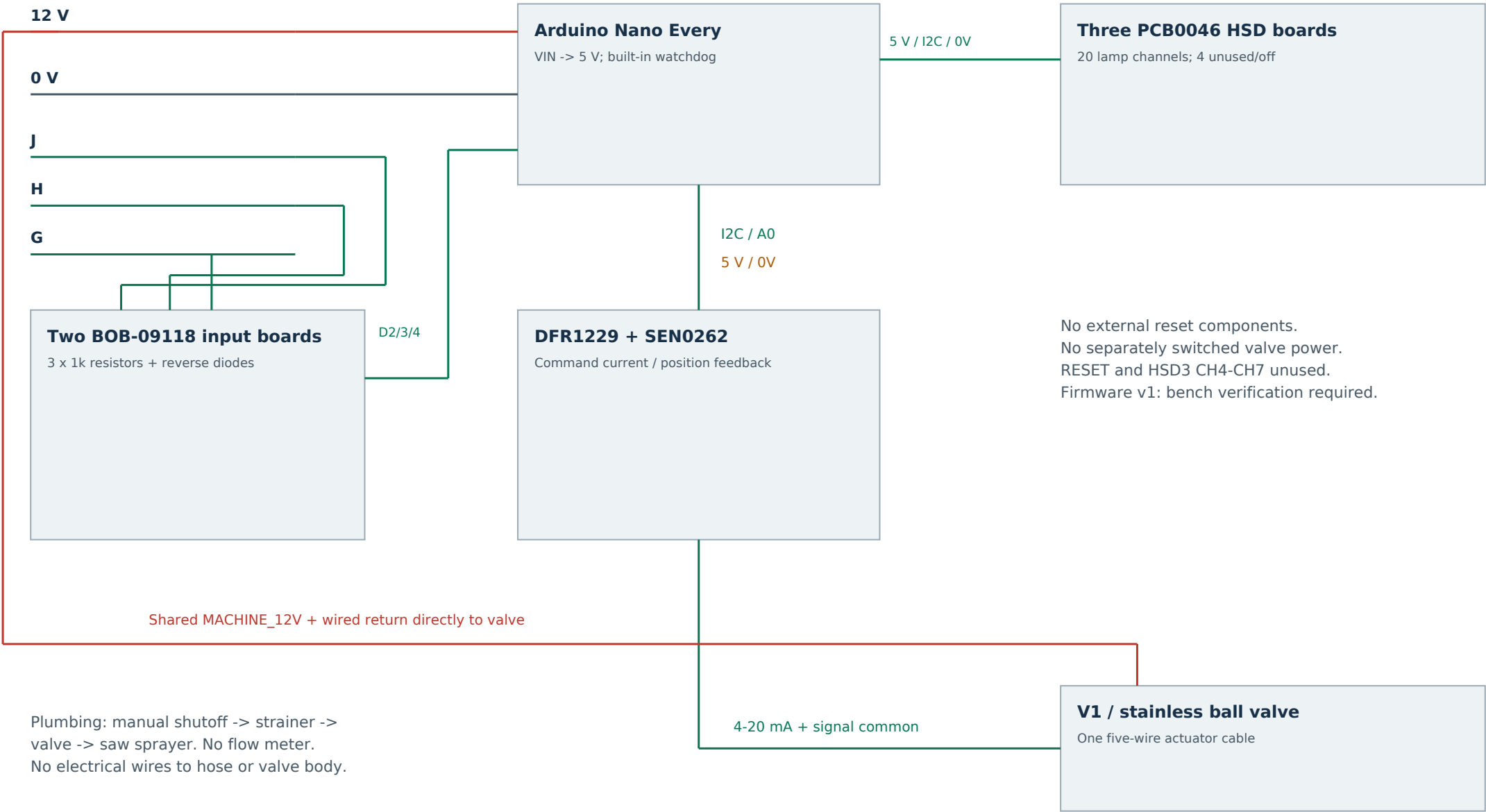
Changes from Revision F

- Remove Q1, R11 and R12 and their six external reset connections. Nano RESET headers remain unconnected.
- Keep the ATmega4809 internal watchdog. HSD3 CH4-CH7 stay off and unconnected. No extra watchdog board is added.
- Show five individual incoming wires from the left: 12 V, 0 V, J, H, G. IN denotes this drawing boundary; the reused connector is not drawn as a harness block.
- Preserve all 102 remaining wire IDs and all retained circuit values. The drawing has 38 references, including the incoming-wire boundary, and 37 physical component/module references.
- Supply real v1 source, native automated tests and a Nano Every compile workflow. This is a bench release, not proof of field reliability.

ROCK-SAW WATER CONTROL / REVISED ELECTRICAL OVERVIEW

Detailed terminal circuits: sheets 01-06. Every external wire is listed in connections.csv.

INCOMING MACHINE WIRES



OVERVIEW ONLY: grouped links represent multiple conductors. Use the single-page drawing to assemble.

REV G | 08 OCT 2026 | BENCH DESIGN - H1-H5 HOLD | Named nets connect across sheets

Solid dot = joined wires. Crossing without a dot = no connection. Boxes show functions, not physical footprints.

01 / POWER AND LOCAL BUS

Supply domains, module grounds, I2C addresses and Nano connections

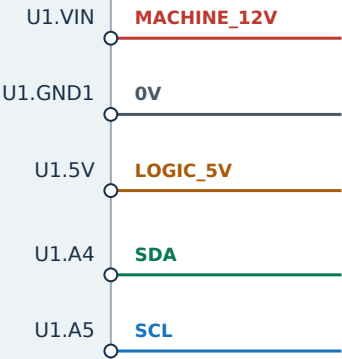
INCOMING MACHINE WIRES



IN.J / IN.H / IN.G are named input wires.
Actual machine cavities: H1 HOLD.
No additional fuse block; H2 HOLD.

U1 / Nano Every

Only VIN accepts machine voltage



PARALLEL SUPPLY CONNECTIONS

MACHINE_12V -> TB12 -> U1 VIN + HSD1/2/3 VIN
0V -> TB0 -> all power/signal returns
LOGIC_5V -> TB5 -> U2 VCC, U3 VCC, HSD V,
O1/O2 HV (no flow meter)

HSD load GND also grounds its logic internally.
Do not carry lamp current through a logic header.
U2 OUT_GND = U2 GND internally.
U1 GND2 = U1 GND1 internally.
V1 RED also uses MACHINE_12V; no relay supply.

LOCAL I2C - BELOW 100 kHz

SDA: U1 A4 -> U2 SDA + HSD1/2/3 D
SCL: U1 A5 -> U2 SCL + HSD1/2/3 C
U2: 0x58; HSD pairs: 0x60/61, 0x62/63, 0x64/65
Use U2 onboard 4.7k pull-ups only.
Open all HSD SDA/SCL pull-up jumpers.
All bus wiring remains inside enclosure.

Local HSD bypass at each VIN/GND: 47 uF / 35 V (+ to VIN) in parallel with 100 nF / 50 V.
USB can energize the logic rail. Disconnect machine, valve and water before USB servicing.

02 / G, H AND J INPUTS

SparkFun BOB-09118 v1.2: HV terminal receives 5 V, not machine voltage



O1/O2 HV -> LOGIC_5V; HV-GND and INPUT_GND -> 0V. O2 IN2 -> INPUT_GND.
O2 OUT2 and both JP2 pin 1 pads unused. HIGH machine input -> HIGH Arduino input.
The three diodes are across the input LED circuits, after the external series resistors.

03 / DIRECT VALVE CONNECTIONS

Shared supply; internal Nano watchdog only; no external reset or valve-power switch

U2 / DFR1229

VCC 5 V; GND 0V; SDA A4; SCL A5
OUT -> valve GREEN directly.
OUT_GND = GND -> WHITE / 0V.
Initialize to 4 mA CLOSED before controls.
0 mA is NOT the documented closed command.

U3 / SEN0262

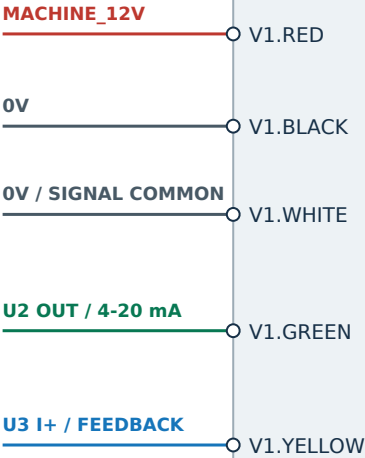
VCC 5 V; GND 0V; SIGNAL -> R4 / A0
I+ -> valve YELLOW directly.
I- -> WHITE / 0V. No added shunt.

CONTROLLER RECOVERY / INTERNAL WATCHDOG

ATmega4809 built-in watchdog: nominal 2.048 s.
No external reset circuit, relay, or watchdog board.
Nano RESET headers and HSD3 CH4-CH7 remain unconnected.
A restart requests CLOSED; it cannot guarantee water shutoff.

V1 / USS-MSV50030

One five-wire actuator cable



Valve power remains connected.
If recovery fails: stop work, shut off water,
then unplug/replug the machine connector.
No restored ON state after reboot.

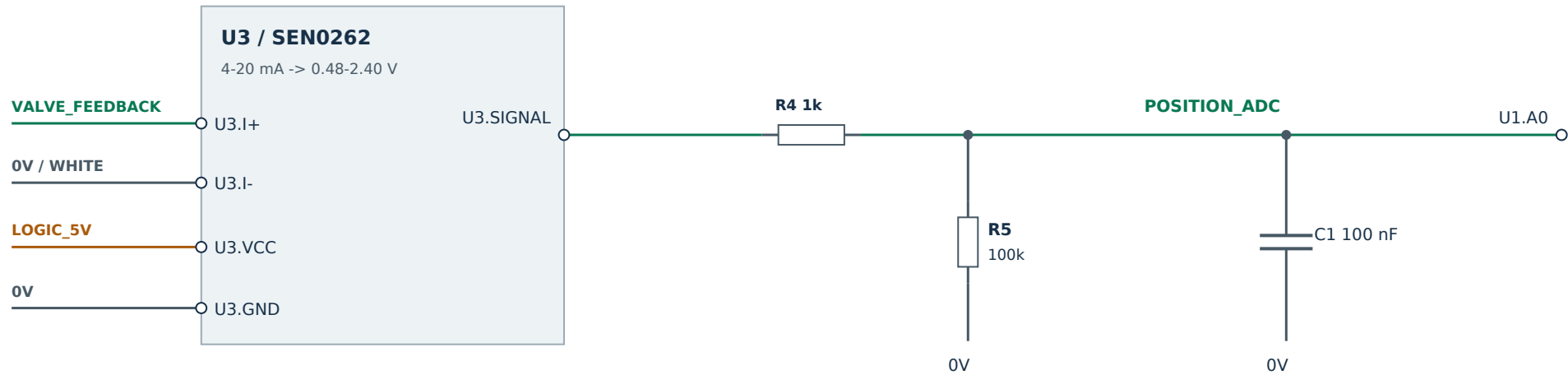
No independent valve-power cut or actuator-current measurement. Reset does not guarantee closure.
Disconnect valve and machine before USB-only service. Verify power sequencing on the bench.

REV G | 08 OCT 2026 | BENCH DESIGN - H1-H5 HOLD | Named nets connect across sheets

Solid dot = joined wires. Crossing without a dot = no connection. Boxes show functions, not physical footprints.

04 / ANALOG POSITION FEEDBACK

No 12 V connection anywhere on this sheet; U2 current-loop output is not an ADC signal



R5 biases an open SIGNAL wire low. R4/R5 reduce the measured voltage by about 1%; calibrate.
No extra 250-ohm resistor across U3 I+/I-. C1 and R5 return locally to the Nano ground.

Revision G omits the flow meter and its entire interface.
Nano D8 and D9 are unused; R6-R9 and C2 are not installed.

Valve position feedback remains required for normal control and recovery.
An indicated closed position is not an independent measurement of stopped water.

Optional manual calibration: collect all nozzle output at known valve positions.
Store a validated monotone flow/position table; there is no automatic meter sweep.

05 / LAMP OUTPUTS AND BOARD CONFIGURATION

Use the printed channel numbers. Each blue/white lamp needs TWO separate switched positives.

HSD1 / PCB0046 V2
0x60/61 | LOAD VIN 12 V; VDD 5 V

CH0

L1 BLUE +

CH1

L1 WHITE +

CH2

L2 BLUE +

CH3

L2 WHITE +

CH4

L3 BLUE +

CH5

L3 WHITE +

CH6

L4 BLUE +

CH7

L4 WHITE +

HSD2 / PCB0046 V2
0x62/63 | LOAD VIN 12 V; VDD 5 V

CH0

L5 BLUE +

CH1

L5 WHITE +

CH2

L6 BLUE +

CH3

L6 WHITE +

CH4

L7 BLUE +

CH5

L7 WHITE +

CH6

L8 BLUE +

CH7

L8 WHITE +

HSD3 / PCB0046 V2
0x64/65 | LOAD VIN 12 V; VDD 5 V

CH0

L9 BLUE +

CH1

L9 WHITE +

CH2

L10 BLUE +

CH3

L10 WHITE +

CH4

UNUSED / OFF

CH5

UNUSED / OFF

CH6

UNUSED / OFF

CH7

UNUSED / OFF

VIN -> MACHINE_12V; LOAD GND -> 0V
D -> SDA; C -> SCL; V -> LOGIC_5V
All lamp BLACK leads -> rated 0V return

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All HSD SDA/SCL pull-up jumpers OPEN. Address: HSD1 all open; HSD2 A1 closed; HSD3 A2 closed.
V2 component-side: printed ch7 at top, ch0 at bottom; schematic OUT1 = ch7, OUT8 = ch0.
Top supply: VIN left / G right. Bottom supply: G left / VIN right. Verify markings before power.

06 / TERMINAL ORIENTATION AND ASSEMBLY NOTES

Physical pin references apply ONLY to the stated package and revision

LOW-VOLTAGE CONNECTIONS

Nano D7, D8 and D9 are unused; leave unconnected.
No external temperature probes or flow meter.

Retain valve feedback through U3 to Nano A0.
Position feedback does not measure actual water flow.

C1: at A0 / logic return.
Nano RESET headers: leave unconnected.
C7-C12: local HSD load-supply bypass.
No Q1, R11, R12 or external watchdog circuit.
Removed reference numbers are not reused.

POWER AND DIAGNOSTICS

Valve and electronics share MACHINE_12V.
No relay or separate coil regulator.
All load returns go to rated ground distribution.
Keep lamp current out of the Nano/ADC return.

Driver thermal protection remains built in.
Generic driver FAULT does not measure ambient heat.
THER is a control input, not a temperature output.
Codes 2 and 9: no corresponding hardware sensors.

IN labels name wires, not machine pin cavities.

FIVE INCOMING WIRES

12 V / 0 V / J / H / G enter from the left.
No connector or harness box is shown.
IN is a drawing boundary, not a purchased component.
Verify the actual reused connector before wiring.
G -> D2; H -> D3; J -> D4 through O1/O2.

WATCHDOG CONFIGURATION

Nano internal hardware watchdog: nominal 2.048 s.
Feed only after a bounded foreground loop completes.
Every I2C transaction has a 20 ms time limit.
Reinitialize DAC and lamp boards after restart.
Keep HSD3 CH4-CH7 off and unconnected.
Verify recovery on actual hardware before field use.

Component roles

Reference	Component	Function
U1	Genuine Arduino Nano Every ABX00028	Main controller, onboard 5 V regulator and internal hardware watchdog
U2	DFRobot DFR1229 / GP8600	Converts digital commands to valve control current
U3	DFRobot SEN0262	Converts reported valve position current to voltage for A0
HSD1-3	Serial Wombat PCB0046 HSD V2	Twenty lamp outputs; HSD3 CH4-CH7 unused/off
O1/O2	SparkFun BOB-09118 v1.2	Three 12 V input channels converted to active-high 5 V logic
V1	U.S. Solid USS-MSV50030	1/2-inch stainless five-wire proportional valve
L1-10	Nilight TL-248BW or verified equivalent	Common negative, separate BLUE and WHITE positive inputs

Power and local wiring

The reused machine connector supplies MACHINE_12V and 0V. TB12 feeds U1 VIN, HSD1/2/3 VIN and V1 RED in parallel. TB0 collects lamp, actuator and electronic returns. Keep load current out of Nano headers and signal-ground wiring. HSD logic G, load grounds and channel-return terminals are internally common.

The Nano onboard switching regulator supplies TB5 and the logic rails of U2, U3, all three HSDs, and O1/O2 HV. HV on the SparkFun input board means its logic-side voltage: use 5 V, not 12 V. Do not add a second 5 V source. Verify total load and regulator temperature in the enclosure.

Retain one 47 uF / 35 V / 105 C electrolytic and one 100 nF / 50 V X7R ceramic across each HSD VIN and LOAD_GND: C7/C10, C8/C11, C9/C12. Electrolytic positive goes to VIN. Place close to each board with short leads. These are local supply buffers, not comprehensive surge protection.

The 9-16 V envelope used in calculations is a proposed bench range, not a fully qualified supply range. V1 requires at least 9 V at its leads. Existing upstream fuse protection and wiring must be checked together.

G/H/J inputs and main-power-off behavior

IN.G, IN.H and IN.J identify the three incoming control wires. Each enters a 1 kohm / 0.5 W series resistor (R1-R3), followed by a BOB-09118 input and reverse diode (D1-D3). Diode cathode/band faces the board input; anode connects to its INPUT_GND. O1 OUT1/OUT2 connect to Nano D2/D3; O2 OUT1 connects to D4. O2 IN2 is grounded and OUT2 is unused.

The selected board includes 220 ohms in its input LED path. At 12 V, estimated asserted-input draw is $(12 - 1.2) / 1220 = 8.9$ mA. About 0.21 Ah is consumed by one continuously asserted input over 24 hours. A brief button press uses very little charge; the machine's own circuit consumption is additional.

With main supply absent, ground retained and USB unplugged, G/H/J energize only the optocoupler input side. Its output pull-ups use controller 5 V, so the specified circuit has no conductive positive-input path into the logic supply. Confirm this with off-state rail measurements. Shared grounds mean this is not complete galvanic isolation of the assembly.

At power-up, request closed, run the one-second-white / one-second-blue lamp test without accepting gestures, then require released J and centered G/H for 100 ms. Discard inputs held through startup. After fault acknowledgement, held G/H must not prevent recovery; consume the old rocker action and require a fresh J press.

Valve command and feedback

Valve lead	Revision G connection
RED	TB12 MACHINE_12V directly
BLACK	TB0 power return
GREEN	U2 OUT directly
WHITE	TB0 signal common, U2 OUT_GND and U3 I-; verify common compatibility
YELLOW	U3 I+ directly

Initialize U2 to its 0-20 mA range, then command 4 mA for closed. The module is not intrinsically restricted to 4-20 mA: code zero produces 0 mA, which the valve manual does not define as closed. Nominal 16-bit codes are 13107 (closed), 39321 (50% travel), and 65535 (full travel). These are valve-position percentages, not calibrated flow percentages.

U2 contains a boosted output stage; never connect OUT to an Arduino input. Test it first with a separate removable 250-ohm precision load, then the actual valve. Its required input burden voltage and loss-of-signal behavior require measurement; matching the words 4-20 mA is not enough to establish the complete interface.

U3 uses a nominal 120-ohm conversion: 4-20 mA gives 0.48-2.40 V. SIGNAL connects through $R4 = 1\text{ kohm}$ to A0. $R5 = 100\text{ kohm}$ and $C1 = 100\text{ nF}$ go from A0 to local Nano ground. The resistor divider gives a nominal 100/101 attenuation: approximately 0.475-2.376 V at A0. Calibrate with the measured 5 V reference. No additional parallel shunt is installed. This input is not galvanically isolated and its resistor/filter does not make it 12 V tolerant.

Valve-reported position is not measured flow or proof that water has stopped. Bucket/stopwatch calibration through the actual nozzles can provide a monotone position/flow table. No automatic meter sweep or live GPM is provided.

Shared-power and USB limitation

Removing the relay removes physical isolation of the signal positives. Shared machine power avoids intentionally leaving the command module energized while switching off the valve alone. It does not prove harmless behavior during brownouts, unplugged valve power, or unequal rail decay. Measure these states during H4.

USB can independently energize the Nano 5 V rail and U2/U3. Before USB-only programming or testing, unplug the machine connector and disconnect the valve cable; isolate the water. Do not connect USB with the unpowered valve's signal leads still attached. There is no signal-disconnect circuit.

Internal watchdog only

Firmware v1 enables the Nano ATmega4809 internal hardware watchdog at a nominal 2.048 seconds. It is fed only after a bounded foreground loop completes. Software I2C transactions are limited to 20 ms each; serial output is sent in short nonblocking pieces. A handled fault remains a valid loop outcome and does not deliberately create a reboot loop.

A watchdog restart records the reset cause, requests 4 mA CLOSED before board initialization, and never restores ON or Flush. DAC and Wombat outputs can retain their previous state during a Nano reset; initialization explicitly reconfigures them. The internal watchdog cannot reset a failed peripheral. Firmware retries bounded bus recovery and board initialization while retaining the fault latch. If recovery fails, manually shut off water and power-cycle the shared supply.

Nano RESET headers and HSD3 CH4-CH7 have no external connections. All four spare outputs are initialized off; only CH0-CH3 of HSD3 appear in lamp writes. No Wombat host-reset mode is configured. The internal watchdog, brownout behavior and actual peripheral recovery still require H5 tests. A reset or power removal is not a guaranteed valve closure.

Faults and detection limits

Revision G has no independently switchable actuator supply and no actuator-current sensor. Do not claim to detect actuator overcurrent with a lamp-driver board whose outputs no longer supply that actuator. Code 2 has no fitted hardware detector; code 9 remains reserved because there are no external temperature probes. Wombat current and thermal diagnostics apply to its lamp outputs only.

Hardware firmware must use best-effort closed commands when a fault occurs and the command path remains usable, then check reported position within a bounded deadline. If the command or feedback path is broken, report closure as unknown. Do not claim that an inhibit flag stopped the valve motor or water: there is no independent power cut. A persistent jam or uncertain closure requires the upstream manual water shutoff and repair. Fault acknowledgement must not automatically restore water ON or Flush.

The existing browser simulator retains its tested gesture and fault-display scenarios, including simulated faults without fitted detectors. Its movement-inhibit model is not a validated physical response for this relay-free design. The physical v1 response is documented separately; the simulator is not Arduino firmware or evidence of valve shutdown. Keep this limitation visible rather than silently claiming identical fault coverage.

Logging, enclosure and service

Firmware v1 keeps a 16-event RAM ring and two CRC-checked, alternating 24-byte EEPROM records containing settings, latched fault mask and reset flags. The commit marker is written last; native tests interrupt every write boundary. Normal setting saves wait two seconds; fault changes request an immediate transaction, which still needs multiple loop passes. Sudden power loss can lose the latest event. Serial status, log and boards commands are read-only. RAM history is lost on power removal; this is not a permanent event recorder. ON and Flush are never stored.

Prefer a gasketed enclosure, sealed glands, supported wiring and removable modules. Conformal coating can protect clean boards while leaving connectors/service points accessible. The Wombat MOSFET outputs reduce load-switching dissipation compared with Darlington drivers; whole-board temperature still needs checking. Do not claim a potting rating for these assembled boards. Permanent resin conflicts with inexpensive replacement.

Practical bench sequence

1. Inspect every wire and polarity with power absent. Confirm no 12 V connection to 5 V, A0, RESET or G/H/J logic pins. Check capacitor polarity.
2. Power the Nano and logic modules with valve disconnected. Measure the 5 V rail and module load. Check USB-only service separation.
3. Exercise all three machine inputs, including asserted inputs with main supply off, then power restoration while held.
4. Test one lamp color, then the complete white/blue startup test and animations. Check current, temperature, return wiring and that HSD3 CH4-CH7 remain off.
5. Test internal watchdog recovery with valve/water disconnected; confirm boot requests closed, restores neither ON nor Flush, and reports watchdog reset. Interrupt the bus and power-cycle a Wombat; confirm bounded retries and retained fault indication.
6. Test U2 on the removable load and U3 with known currents. Then verify actual valve wiring, common returns, command range and reported position without pressurized water.
7. Test shared-power sequencing, broken power/signal connections and bounded communication failures. Confirm boot always commands closed; identify any condition requiring manual water shutoff.
8. Run water through the actual nozzles. Verify settings, Max Cap and Flush behavior, calibration, manual shutoff, enclosure heat and recovery after interruptions. Record actual evidence in Validation.

Source register

Reviewed 8 October 2026. Source inspection is not physical qualification.

Source	Used for
Arduino Nano Every schematic	VIN regulator, shared RESET headers, reset pull-up/capacitor, USB power
ATmega4808/4809 datasheet	Internal watchdog, reset flags and timing
Serial Wombat PCB0046	Channels, addresses and diagnostic limits
TI TPS4H160-Q1	HSD output leakage, protected MOSFET switching
SparkFun input schematic	Input resistor/LED/inverter topology
U.S. Solid valve manual	Five-wire connections and 4-20 mA position command
DFRobot DFR1229	Current command configuration
DFRobot SEN0262	Position-current conversion

Complete external connection schedule

Every row is a conductor or local soldered connection. Repeated distribution terminals denote joined bus terminals, not a bundle forced into one screw. Module-internal common connections are listed separately. Wire gauges and protection remain H2.

Wire	Net	From	To
W001	MACHINE_12V	TB12.rail	IN.12V
W002	MACHINE_12V	TB12.rail	U1.VIN
W003	MACHINE_12V	TB12.rail	HSD1.VIN
W004	MACHINE_12V	TB12.rail	HSD2.VIN
W005	MACHINE_12V	TB12.rail	HSD3.VIN
W006	0V	TB0.rail	IN.0V
W007	0V	TB0.rail	U1.GND1
W008	0V	TB0.rail	U2.GND
W009	0V	TB0.rail	U3.GND
W010	0V	TB0.rail	HSD1.LOAD_GND
W011	0V	TB0.rail	HSD2.LOAD_GND
W012	0V	TB0.rail	HSD3.LOAD_GND
W013	0V	TB0.rail	O1.INPUT_GND
W014	0V	TB0.rail	O1.HV_GND
W015	0V	TB0.rail	O2.INPUT_GND
W016	0V	TB0.rail	O2.HV_GND
W017	0V	TB0.rail	V1.BLACK
W018	0V	TB0.rail	V1.WHITE
W019	0V	TB0.rail	U3.I-
W023	LOGIC_5V	TB5.rail	U1.5V
W024	LOGIC_5V	TB5.rail	U2.VCC
W025	LOGIC_5V	TB5.rail	U3.VCC
W026	LOGIC_5V	TB5.rail	HSD1.V
W027	LOGIC_5V	TB5.rail	HSD2.V
W028	LOGIC_5V	TB5.rail	HSD3.V
W029	LOGIC_5V	TB5.rail	O1.HV
W030	LOGIC_5V	TB5.rail	O2.HV
W032	SDA	U1.A4	U2.SDA
W033	SDA	U1.A4	HSD1.D

Wire	Net	From	To
W034	SDA	U1.A4	HSD2.D
W035	SDA	U1.A4	HSD3.D
W036	SCL	U1.A5	U2.SCL
W037	SCL	U1.A5	HSD1.C
W038	SCL	U1.A5	HSD2.C
W039	SCL	U1.A5	HSD3.C
W040	MACHINE_G	IN.G	R1.1
W041	OPTO_G	R1.2	O1.IN1
W042	OPTO_G	R1.2	D1.K
W043	0V	O1.INPUT_GND	D1.A
W044	G_LOGIC	O1.OUT1	U1.D2
W045	MACHINE_H	IN.H	R2.1
W046	OPTO_H	R2.2	O1.IN2
W047	OPTO_H	R2.2	D2.K
W048	0V	O1.INPUT_GND	D2.A
W049	H_LOGIC	O1.OUT2	U1.D3
W050	MACHINE_J	IN.J	R3.1
W051	OPTO_J	R3.2	O2.IN1
W052	OPTO_J	R3.2	D3.K
W053	0V	O2.INPUT_GND	D3.A
W054	J_LOGIC	O2.OUT1	U1.D4
W055	0V	O2.INPUT_GND	O2.IN2
W056	L1_BLUE	HSD1.CH0	L1.BLUE
W057	L1_WHITE	HSD1.CH1	L1.WHITE
W058	0V	TB0.rail	L1.BLACK
W059	L2_BLUE	HSD1.CH2	L2.BLUE
W060	L2_WHITE	HSD1.CH3	L2.WHITE
W061	0V	TB0.rail	L2.BLACK
W062	L3_BLUE	HSD1.CH4	L3.BLUE
W063	L3_WHITE	HSD1.CH5	L3.WHITE
W064	0V	TB0.rail	L3.BLACK
W065	L4_BLUE	HSD1.CH6	L4.BLUE

Wire	Net	From	To
W066	L4_WHITE	HSD1.CH7	L4.WHITE
W067	0V	TB0.rail	L4.BLACK
W068	L5_BLUE	HSD2.CH0	L5.BLUE
W069	L5_WHITE	HSD2.CH1	L5.WHITE
W070	0V	TB0.rail	L5.BLACK
W071	L6_BLUE	HSD2.CH2	L6.BLUE
W072	L6_WHITE	HSD2.CH3	L6.WHITE
W073	0V	TB0.rail	L6.BLACK
W074	L7_BLUE	HSD2.CH4	L7.BLUE
W075	L7_WHITE	HSD2.CH5	L7.WHITE
W076	0V	TB0.rail	L7.BLACK
W077	L8_BLUE	HSD2.CH6	L8.BLUE
W078	L8_WHITE	HSD2.CH7	L8.WHITE
W079	0V	TB0.rail	L8.BLACK
W080	L9_BLUE	HSD3.CH0	L9.BLUE
W081	L9_WHITE	HSD3.CH1	L9.WHITE
W082	0V	TB0.rail	L9.BLACK
W083	L10_BLUE	HSD3.CH2	L10.BLUE
W084	L10_WHITE	HSD3.CH3	L10.WHITE
W085	0V	TB0.rail	L10.BLACK
W086	MACHINE_12V	TB12.rail	V1.RED
W095	VALVE_COMMAND	U2.OUT	V1.GREEN
W097	VALVE_FEEDBACK	V1.YELLOW	U3.I+
W099	POSITION_RAW	U3.SIGNAL	R4.1
W100	POSITION_ADC	U1.A0	R4.2
W101	POSITION_ADC	U1.A0	R5.1
W102	POSITION_ADC	U1.A0	C1.1
W103	0V	U1.GND1	R5.2
W104	0V	U1.GND1	C1.2
W115	MACHINE_12V	HSD1.VIN	C7.+
W116	MACHINE_12V	HSD1.VIN	C10.1
W117	0V	HSD1.LOAD_GND	C7.-

Wire	Net	From	To
W118	0V	HSD1.LOAD_GND	C10.2
W119	MACHINE_12V	HSD2.VIN	C8.+
W120	MACHINE_12V	HSD2.VIN	C11.1
W121	0V	HSD2.LOAD_GND	C8.-
W122	0V	HSD2.LOAD_GND	C11.2
W123	MACHINE_12V	HSD3.VIN	C9.+
W124	MACHINE_12V	HSD3.VIN	C12.1
W125	0V	HSD3.LOAD_GND	C9.-
W126	0V	HSD3.LOAD_GND	C12.2

Internal connections - no added wire

HSD1.LOAD_GND = HSD1.G; HSD1.LOAD_GND = HSD1.G0; HSD1.LOAD_GND = HSD1.G1; HSD1.LOAD_GND = HSD1.G2; HSD1.LOAD_GND = HSD1.G3; HSD1.LOAD_GND = HSD1.G4; HSD1.LOAD_GND = HSD1.G5; HSD1.LOAD_GND = HSD1.G6; HSD1.LOAD_GND = HSD1.G7; HSD2.LOAD_GND = HSD2.G; HSD2.LOAD_GND = HSD2.G0; HSD2.LOAD_GND = HSD2.G1; HSD2.LOAD_GND = HSD2.G2; HSD2.LOAD_GND = HSD2.G3; HSD2.LOAD_GND = HSD2.G4; HSD2.LOAD_GND = HSD2.G5; HSD2.LOAD_GND = HSD2.G6; HSD2.LOAD_GND = HSD2.G7; HSD3.LOAD_GND = HSD3.G; HSD3.LOAD_GND = HSD3.G0; HSD3.LOAD_GND = HSD3.G1; HSD3.LOAD_GND = HSD3.G2; HSD3.LOAD_GND = HSD3.G3; HSD3.LOAD_GND = HSD3.G4; HSD3.LOAD_GND = HSD3.G5; HSD3.LOAD_GND = HSD3.G6; HSD3.LOAD_GND = HSD3.G7; U1.GND1 = U1.GND2; U2.GND = U2.OUT_GND

Unused external terminals

O1.JP2_1; O2.JP2_1; O2.OUT2; HSD3.CH5; HSD3.CH6; HSD3.CH7; U1.3V3; U1.A1; U1.A2; U1.A3; U1.A6; U1.A7; U1.AREF; U1.RX; U1.TX; U1.D5; U1.D6; U1.D7; U1.D10; U1.D11; U1.D12; U1.D13; U1.D8; U1.D9; U1.RESET1; U1.RESET2; HSD3.CH4

Complete parts schedule

Values specify the proposed circuit. Distribution hardware, cable sizes, sealed connectors and enclosure remain subject to physical hold points. Test instruments and the removable DAC test resistor are not installed circuit components.

Ref	Specification	Note
TB12	12 V distribution	Joined terminals; hardware and wire rating subject to H2
TB5	Logic 5 V distribution	Joined terminals; hardware and wire rating subject to H2
TB0	Ground distribution	Joined terminals; hardware and wire rating subject to H2
U1	Arduino Nano Every ABX00028	
U2	DFRobot DFR1229	
U3	DFRobot SEN0262	
HSD1	Serial Wombat PCB0046 HSD V2	
HSD2	Serial Wombat PCB0046 HSD V2	
HSD3	Serial Wombat PCB0046 HSD V2	
O1	SparkFun BOB-09118 v1.2	
O2	SparkFun BOB-09118 v1.2	
V1	U.S. Solid USS-MSV50030	
L1	Nilight TL-248BW	
L2	Nilight TL-248BW	
L3	Nilight TL-248BW	
L4	Nilight TL-248BW	
L5	Nilight TL-248BW	
L6	Nilight TL-248BW	
L7	Nilight TL-248BW	
L8	Nilight TL-248BW	
L9	Nilight TL-248BW	
L10	Nilight TL-248BW	
R1	1000 ohm 1% 0.5 W	
D1	1N4148, axial	Band = K / cathode
R2	1000 ohm 1% 0.5 W	
D2	1N4148, axial	Band = K / cathode
R3	1000 ohm 1% 0.5 W	

Ref	Specification	Note
D3	1N4148, axial	Band = K / cathode
R4	1000 ohm 1% 0.25 W	
R5	100000 ohm 1% 0.25 W	
C1	100 nF X7R 50 V	
C10	100 nF X7R 50 V	
C11	100 nF X7R 50 V	
C12	100 nF X7R 50 V	
C7	47 uF 35 V 105 C electrolytic	
C8	47 uF 35 V 105 C electrolytic	
C9	47 uF 35 V 105 C electrolytic	
IN	Five incoming machine wires	No connector symbol; map these signals to the existing machine wiring